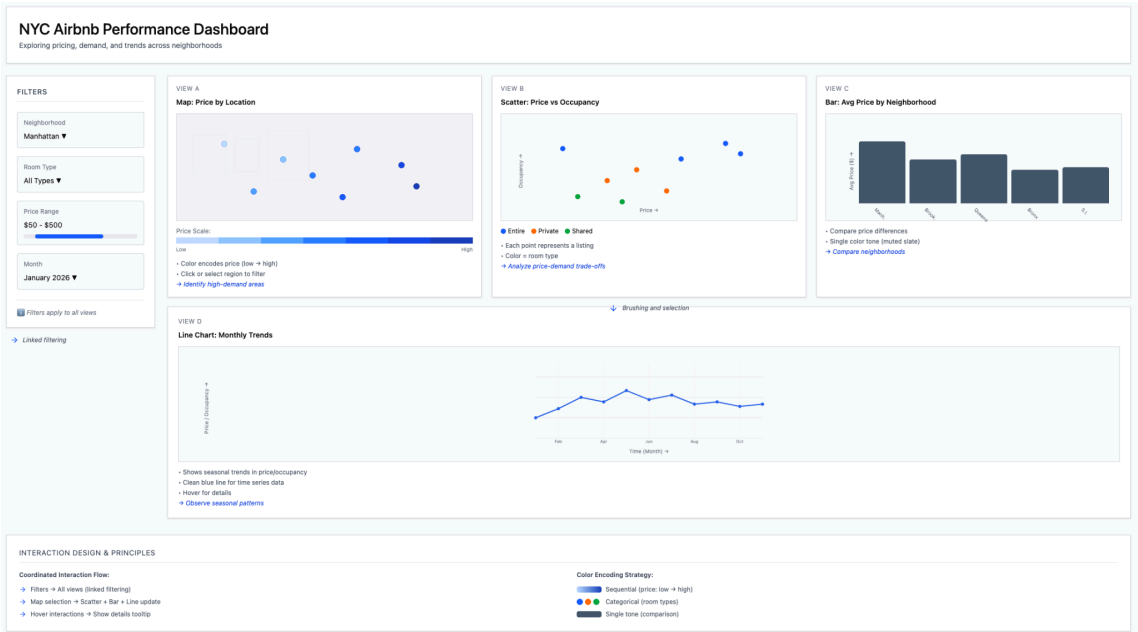
Visualizing Spatial and Temporal Patterns in NYC Airbnb Performance Using an Interactive Multi-View Dashboard

Understanding the interplay of price, demand, location, and time in short-term rental markets in urban environments such as New York City is challenging because traditional statistical summaries fail to reveal spatial and temporal patterns. This project proposes the design of an interactive visualization system that facilitates data exploration through coordinated views. A publicly available dataset from Kaggle is used, consisting of 4,326 Airbnb listings with monthly performance data from 2021 to 2026. Key attributes include geographic coordinates, average daily rate, occupancy rate, revenue, property characteristics, and temporal information. The combination of spatial, temporal, and multivariate data introduces challenges in effectively encoding multiple variables while maintaining clarity and enabling meaningful comparison across views. These challenges require interactive visualization techniques to support exploration, comparison, and pattern discovery.

To address this problem, the project proposes the development of an interactive dashboard built using Python and Plotly Dash. The system includes multiple coordinated views, such as a geographic map, a scatter plot, and a time-series chart. These views enable users to filter and explore subsets of the data interactively. The expected contribution of this project is a visual analytics system that enables users to explore complex relationships between price, demand, and location in a unified interface, revealing patterns that are difficult to identify using traditional analytical methods, such as high-demand regions, pricing inefficiencies, and seasonal trends.



Reference

Kaggle. Airbnb Data: New York City 5Y (2021–2026)
https://www.kaggle.com/datasets/jasonairroi/airbnb-short-term-rental-data-new-york-city